

MedRAG System Verification Test Cases

A standardized set of verification cases including patient profile inputs and expected evidence-based clinical outputs.

I. Summary Table of Test Cases

ID	Test Case Title	Target Clinical Query	Primary Evidence Source
TC-001	Renal Contraindication (Metformin)	Can a patient with diabetes and severe renal impairment take Metformin?	DailyMed / FDA Label
TC-002	Pregnancy Safety Warning (Losartan)	Is Losartan safe to control high blood pressure in a pregnant patient?	DailyMed / FDA Label
TC-003	Dual RAAS Blockade Interaction	Can a patient take Lisinopril and Losartan together to manage high blood pressure?	DailyMed / FDA Label
TC-004	Ethambutol Visual Toxicity & Monitoring	What is the main side effect associated with Ethambutol and what checkups are needed?	Clinical Guidelines
TC-005	First-Line Hypertension Guidelines	What are the first-line drug choices recommended for newly diagnosed Hypertension?	Clinical Guidelines

II. Detailed Patient Profiles & Expected Outputs

TC-001: Renal Contraindication (Metformin)

Clinical Query: "Can a patient with diabetes and severe renal impairment take Metformin?"

Patient Context Parameter	Input Context Value
Age	62 years
Sex	Male
Conditions	Type 2 Diabetes, Chronic Kidney Disease (CKD)
eGFR	25 mL/min/1.73m ² (Severe Impairment)
Current Medications	None
Known Allergies	None

Expected Output Summary: Metformin is strictly contraindicated in this patient. FDA regulations and clinical guidelines specify that Metformin must not be initiated or continued in patients with an eGFR below 30 mL/min/1.73m² due to the significant risk of drug accumulation and associated Lactic Acidosis (a rare but potentially fatal metabolic complication).

Expected Reference Citations to look for:

- 1. Metformin FDA Label - Contraindications Section (eGFR < 30)
- 2. Diabetes Clinical Guidelines (Recommendation to discontinue Metformin)

TC-002: Pregnancy Safety Warning (Losartan)

Clinical Query: "Is Losartan safe to control high blood pressure in a pregnant patient?"

Patient Context Parameter	Input Context Value
Age	29 years
Sex	Female
Conditions	Essential Hypertension
Pregnancy Status	Pregnant (2nd Trimester)
Current Medications	None
Known Allergies	None

Expected Output Summary: Losartan is contraindicated. Angiotensin II Receptor Blockers (ARBs) carry a Boxed Warning for fetal toxicity. Exposure during the second and third trimesters of pregnancy reduces fetal renal function and increases fetal/neonatal morbidity and death.

Expected Reference Citations to look for:

1. Losartan FDA Label - Boxed Warning: Fetal Toxicity
2. Hypertension Guidelines - Pregnant patients should be switched to safe alternatives (e.g. Methyldopa, Labetalol, Nifedipine)

TC-003: Dual RAAS Blockade Interaction

Clinical Query: "Can a patient take Lisinopril and Losartan together to manage high blood pressure?"

Patient Context Parameter	Input Context Value
Age	55 years
Sex	Male
Conditions	Hypertension
Current Medications	Lisinopril 20mg daily
Known Allergies	None
Other Details	Adding Losartan 50mg daily

Expected Output Summary: Dual blockade of the renin-angiotensin-aldosterone system (RAAS) by combining an ACE inhibitor (Lisinopril) and an ARB (Losartan) is not recommended. Clinical studies show no additional therapeutic benefit, but significantly higher rates of hypotension, hyperkalemia, and acute kidney injury.

Expected Reference Citations to look for:

1. Lisinopril and Losartan FDA Labels - Drug Interactions Section
2. Hypertension Guidelines - Recommendation against combining ACEi and ARB therapy

TC-004: Ethambutol Visual Toxicity & Monitoring

Clinical Query: "What is the main side effect associated with Ethambutol and what checkups are needed?"

Patient Context Parameter	Input Context Value
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Age	45 years
Sex	Female
Conditions	Active Tuberculosis (Pulmonary TB)
Current Medications	RIPE regimen (Rifampicin, Isoniazid, Pyrazinamide, Ethambutol)
Known Allergies	None

Expected Output Summary: The primary serious side effect of Ethambutol is Optic Neuritis, which causes decreased visual acuity and red-green color blindness. Patients must undergo baseline and periodic visual testing (Snellen chart and Ishihara color plates) during treatment.

- Expected Reference Citations to look for:**
- 1. Ethambutol FDA Label - Warnings & Precautions (Visual Acuity)
 - 2. Tuberculosis Clinical Guidelines - Monitoring parameters for active pulmonary TB

TC-005: First-Line Hypertension Guidelines

Clinical Query: "What are the first-line drug choices recommended for newly diagnosed Hypertension?"

Patient Context Parameter	Input Context Value
Age	50 years
Sex	Male
Conditions	Essential Hypertension
Current Medications	None (Treatment naive)
Known Allergies	None

Expected Output Summary: First-line pharmacotherapy options include four classes: Thiazide Diuretics (e.g. Chlorthalidone, Hydrochlorothiazide), Calcium Channel Blockers (CCBs, e.g. Amlodipine), ACE Inhibitors (ACEi, e.g. Lisinopril), or Angiotensin Receptor Blockers (ARBs, e.g. Losartan). Selection should account for ethnicity, comorbidities, and age.

- Expected Reference Citations to look for:**
- 1. Hypertension Clinical Practice Guidelines (First-line Recommendations)
 - 2. Amlodipine, Lisinopril, Losartan, Chlorthalidone FDA Labels